

AVOIDANCE OF CONTAMINATED SEDIMENTS BY AN AMPHIPOD (*MELITA PLUMULOSA*), A HARPACTICOID COPEPOD (*NITOCRA SPINIPES*), AND A SNAIL (*PHALLOMEDUSA SOLIDA*)

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Abstract—The distribution of contaminants is seldom homogeneous in aquatic systems. In the present study, the avoidance response of *Melita plumulosa*, *Nitocra spinipes*, and *Phallomedusa solida* when exposed to contaminated sediments was investigated. Test vessels were designed to allow the congruent placement of two sediments and assessment of the movement of organisms between the sediments. When exposed to reference sediment, each species dispersed evenly between test chambers regardless of differences in sediment particle size. In the presence of contaminated sediment, the magnitude and rate of avoidance varied. Avoidance assays showed that test species avoided contaminated sediment as early as 6, 6, and 24 h following exposure for *N. spinipes*, *P. solida*, and *M. plumulosa*, respectively. The 48-h avoidance response of *M. plumulosa* for nine contaminated sediments of varying toxicity showed that avoidance was generally greater for sediments which elicited greater 10-d lethality to this species. The study demonstrated that each of these species has the ability to respond to chemical cues in the environment to inhabit sediment that provides the best opportunity for survival. The avoidance response for each species indicates the potential for developing rapid screening methods to assess sediment quality. Evidence suggests that avoidance was related to sediment toxicity and that static 10-d toxicity methods are likely to overestimate toxicity for species, which would avoid contamination in heterogeneous field settings. Environ. Toxicol. Chem. 2013;32:644–652. © 2012 SETAC

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INTRODUCTION

Environmental contaminants are known to have adverse effects on aquatic organisms. They impact on survival [1] and affect crucial behaviors such as the ability to effectively forage for food, avoid predation, find suitable habitat, compete for resources, and tolerate a number of biotic and abiotic interactions within the surrounding environment [2]. Contaminant-induced behavioral changes can be linked to biochemical and physiological effects at the organism level, which can have flow-on effects at the population level [3]. Therefore, significant changes to crucial behavioral traits of an organism (e.g., foraging behavior, social interaction, and locomotion) may pose a threat to the survival of a population [4,5]. Behavioral traits can be exploited as toxicity test endpoints and, in some cases, have been demonstrated to be sensitive to contaminant exposure [6,7]. Consequently, behavioral endpoints are increasingly being considered to have high ecological relevance as contaminant-induced changes to an organism's behavior offer insight into potentially major environmental impacts [3]. Previous studies have reported contaminant-induced changes to a number of key behaviors, including predator–prey relationships [8], swimming and mobility [5], and contaminant avoidance [7,9].

The concentration of toxicants in surface sediments will often vary spatially, and a number of organisms have demonstrated the ability to detect and actively avoid areas of contamination [10,11]. In the natural environment, individual organisms, and eventually populations, will migrate away from a contaminated site before the exposure results in the uptake of a harmful dose (sublethal or lethal). Hence, avoidance behavior is

a significant factor in determining the extent of exposure [4,6], the magnitude of the hazard, and the overall risk the sediment poses to ecosystem health. Avoidance is often a rapid and easily measurable response, with some suggesting that it is more sensitive than commonly used lethal and sublethal endpoints [6,7]. Behavioral endpoints, such as avoidance, also have the advantage of providing an early warning to potential contamination which may result in lethal impacts [12]. Avoidance behavior has been investigated for a diverse range of aquatic organisms including fish [13], amphipods [9], cladocerans [7], aquatic snails [6], and bivalves [14]. These studies have all reported the usefulness of contaminant avoidance as a behavioral ecotoxicological endpoint.

Contaminant avoidance is likely to be a defense mechanism against the effect of poor environmental conditions, eliminating the threat to survival caused by the presence of toxicants [6]. Contaminant avoidance may also be an adaptive behavior [15]. Lefcort et al. [6] observed that aquatic snails collected from contaminated environments could avoid contaminated sediment, while those from reference sites could not, inferring a potential genetic influence on avoidance behavior. Regardless of the mechanisms triggering the avoidance response, it is important to note that the relocation of a population of aquatic organisms may ensure their survival at a new site. On an ecological scale, the disappearance of the species from the previous location is equivalent to the loss of the entire population [5,7]. Consequently, large areas of sediments that are avoided due to contamination may be classified as high risk, although short-term effects to many species used for toxicity tests may not be observed.

Static toxicity tests rarely reflect the dynamic and heterogeneous natural environment [6] and, therefore, do not always adequately mimic an organism's exposure to contaminants, meaning that these methods may lack ecological significance [1,16]. Despite this, many researchers and regulatory guidelines

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rely on standard methods which confine test organisms to a contaminant, forcing a continuous static exposure [17,18]. The relevance of these common ecotoxicological methods is widely debated in the literature, with chronic test methods often preferred over acute toxicity testing [1,7]. Chronic toxicity tests are considered to offer greater ecological relevance, protection at the population level, increased sensitivity, better prediction of toxicity, and the ability to model population effects [19]. However, chronic toxicity tests generally use long exposure periods that often exceed 20 d [20,21]. Considering the ability of mobile aquatic organisms to avoid contamination, the long test durations often associated with chronic tests do not adequately represent realistic exposure conditions in the natural environment. Understanding the avoidance response of mobile benthic organisms will shed light on the response of benthic invertebrates to contaminated sediments in the environment and allow the development of environmentally relevant methods for assessing sediment toxicity caused by contamination in heterogeneous field settings.

The present study investigated the suitability of contaminant avoidance as an endpoint in whole-sediment toxicity testing using three species of benthic estuarine organisms: an epibenthic amphipod (*Melita plumulosa*), a harpacticoid copepod (*Nitocra spinipes*), and a snail (*Phallomedusa solida*). All of these species are known to graze or ingest sediment particles, making them suitable candidates for sediment toxicity testing [22–24]. For each species the influence of varying physicochemical properties of the sediment on the distribution of test organisms and the optimal exposure time to measure an avoidance response elicited by contaminated sediments were initially determined. This provided optimal methods for avoidance bioassays for assessing the ability of each species to detect and move away from contaminated sediment.

MATERIALS AND METHODS

Test media

Clean seawater was collected from Port Hacking (Sydney, Australia), membrane-filtered (0.45 μm), and acclimated to room temperature ($21 \pm 1^\circ\text{C}$). Where necessary, the salinity of the filtered seawater was adjusted to the test salinity of 30 practical salinity units (psu) using deionized water (18 M Ω /cm; Milli-Q Academic Water System).

Relatively clean silty sediments were collected from an estuarine location in Bonnet Bay on the Woronora River (Sydney, Australia). These sediments had low or negligible concentrations of metal and organic contaminants and have been demonstrated to not cause toxic effects to the organisms used in the study [24,25]. The surface layers (upper 2–4 cm) of sediments were collected using clean Teflon spatulas and press-sieved through a 1.1-mm mesh on-site to remove coarse materials. The sediment was transferred into clean plastic bags with minimal headspace and stored in a cool room at 4°C for no longer than one month. Clean sand that contained negligible contamination [23] was also used as a control material in sediment-avoidance tests. Sediments with 50 and 10% silt were created by mixing the control sediment with clean sand.

Contaminated sediments were chosen to provide exposure to a range of contaminants. Contaminated sediments were collected from estuarine field sites in New South Wales and Tasmania, Australia. Collected sediments were stored at 4°C in the dark, and toxicity testing was undertaken within eight weeks [10,25]. Analyses of physicochemical properties (pH, organic carbon, particle size, acid-volatile sulfide) and metal

contaminants were made on all sediments. Concentrations of total petroleum hydrocarbons were $<250\text{ mg/kg}$ and those of polycyclic aromatic hydrocarbons were $<1\text{ mg/kg}$ in all sediments [10,25,26].

Copper-spiked sediments were prepared using clean silty sediment (collected from the Bonnet Bay reference site) and equilibrated as described previously [27]. The preparation, manipulation, and equilibration of copper-spiked sediments were done in a nitrogen gas-filled glove box at room temperature. Sediments were thoroughly homogenized with a plastic spoon and a bottle roller for 2 h at least two times per week. Sediments were adjusted to pH 7.5 with 1 M NaOH one day after copper spiking and maintained at this pH using small additions of NaOH as required during the one-month equilibration period. Changes in pH, redox potential, and dissolved metals in the porewater were monitored during the equilibration period.

Diesel-spiked sediments were prepared in two stages. Diesel oil was added to suboxic Bonnet Bay sediment to create a 10% diesel stock sediment. Following a two-week equilibration period, an aliquot of stock sediment was combined with suboxic Bonnet Bay sediment to give a final concentration of 5% diesel sediment. Diesel-spiked sediments were maintained in glass containers with minimal headspace. Lids were securely fastened to avoid losses through volatilization. Containers were housed inside two polypropylene bags, which were sealed with duct tape. Both the stock and test sediments were homogenized by vigorous shaking of the containers, followed by 2 to 3 h on a bottle roller at least three times per week during the one-month equilibration period.

Test species

All three species used in the present study were epibenthic deposit feeders found in intertidal estuarine environments of southeastern Australia. *Melita plumulosa* (family Melitidae) is commonly found in estuarine tidal mudflats ranging from silty to sandy sediments in freshwater, estuarine, and marine environments throughout southeastern Australia [28]. Adult specimens of *M. plumulosa* typically range from 8 to 10 mm in length. The harpacticoid copepod species *N. spinipes* (family Ameiridae) is known to adapt to a wide range of environmental conditions (including salinity and temperature) and, as such, has a worldwide distribution [29]. Mature copepods of this species are approximately 400 μm long. *Phallomedusa solida* (family Amphiboloidea), formerly known as *Salinator solida*, is one of the most common deposit-feeding gastropods found to inhabit salt marshes and mangroves in southeastern Australia [22]. Snails of a uniform size ($\sim 10\text{ mm}$ along the ventral surface) were collected for avoidance tests.

Specimens of *M. plumulosa* and *N. spinipes* were obtained from laboratory cultures maintained as per King et al. [28] and Ward et al. [29], respectively. Those of *P. solida* were collected from natural populations occurring in the intertidal mangroves at Bonnet Bay.

Amphipod toxicity testing

Glass beakers and acrylic beaker lids used for toxicity tests were cleaned in a dishwasher (Gallay Scientific) programmed for a phosphate-free detergent wash, a dilute acid wash (1% HNO_3), followed by thorough rinsing with Milli-Q water.

The 10-d whole-sediment toxicity tests with *M. plumulosa* were conducted to assess the effects of continuous exposure to the sediments and were performed in accordance with standard protocols [23]. In brief, tests were performed at $21 \pm 1^\circ\text{C}$ in a

constant environmental chamber (Labec Refrigerated Cycling Incubator) on a 12-h light/12-h dark cycle (light intensity = $3.5 \mu\text{mol photons/s/m}^2$). Dissolved oxygen ($>85\%$), pH (7.5–8.2), salinity (30 ± 1 psu), and temperature ($21 \pm 1^\circ\text{C}$) were monitored and maintained according to Spadaro et al. [23]. For whole-sediment toxicity tests, three replicate 250-ml beakers containing 20 g of test sediment, 200 ml seawater, and 15 adult *M. plumulosa* were used per treatment. Every 2 to 3 d, 80% of the overlying water was replaced with clean seawater. In all tests, water samples were collected before and after each water change for dissolved metal copper analysis. No food was added during the 10-d bioassays.

At the termination of the tests, the contents of each beaker were gently sieved through a $180\text{-}\mu\text{m}$ stainless steel mesh sieve and transferred to large amphipod counting trays. Live amphipods were identified by movement and counted. The sieved sediment was transferred to 120-ml polycarbonate vials with 100 ml of seawater, fixed with 4 ml of 10% v/v neutral phosphate-buffered formalin, and stained with 5 ml of rose bengal solution (0.1 g rose bengal sodium salt [Sigma] per 100 ml Milli-Q). Sediments were left for 72 h to enable any surviving amphipods missed in the initial count to take up the stain and be counted. Tests were considered acceptable if the physicochemical parameters in beakers remained within the limits of pH 7.7 to 8.2, 20 to 22°C , dissolved oxygen $>80\%$ saturation, and salinity 28 to 32 psu throughout the test and if survival of amphipods was on average $\geq 80\%$ in the controls.

Contaminant-avoidance experimental design

Avoidance assays were conducted in test chambers specifically designed and constructed for each test species (Supplemental Data, Fig. S1). Each design utilized a plastic vessel divided into two chambers by a permanent barrier fixed across the bottom of the container. The top of this barrier was at the sediment–water interface and allowed the contiguous placement of two sediments in the same container with minimal mixing of the sediment samples. Removable barriers were constructed from plastic (polyethylene terephthalate plastic film) and used to divide the test chambers from the base of the sediment to the surface of the overlying water. This barrier was used during the initiation and termination of experiments to prevent the migration of test organisms between test chambers. The non-toxic nature of all materials and adhesives was assessed before use to construct the testing apparatus. Test chambers were set up with sediment and filtered seawater and equilibrated for 24 h before the initiation of the experiment. The overlying water was replaced immediately before the test organisms were added.

For *M. plumulosa* and *P. solida*, test chambers were constructed from 3-L polyethylene containers (20×15 cm wide, 10 cm high; Supplemental Data, Fig. S1A). A plastic barrier approximately 1 cm high was adhered to the base across the middle of the container with non-toxic aquarium silicone (Sellys Glass Silicone). Homogenized sediments were placed on the relevant sides of the test chamber, and clean filtered seawater was added to achieve a depth of approximately 8 cm for *M. plumulosa* and approximately 5 cm for *P. solida*. For *P. solida* experiments, gel (Lucas' Pawpaw Ointment) was applied to the chamber walls above the water line to prevent the snails from escaping during testing (this showed no toxic effect in control studies).

The test chambers used for *N. spinipes* were constructed from 1×1 cm wide, 5 cm high polyethylene cells usually used for ultraviolet-visible spectrophotometry (Supplemental Data,

Fig. S1B). An opening (0.5 cm wide $\times$ 2.5 cm tall) was cut into one side of each cell. Two cells were placed together with the opening facing each other and secured in place with adhesive tape. A sediment slurry was created (10 g sediment in 2 ml seawater) and added dropwise to the cells to achieve a layer of sediment approximately 0.5 cm deep overlain by approximately 3.5 cm of filtered seawater. Teflon spatulas were cut down and used to partition the two cells of the test chamber.

Contaminant-avoidance assays

Avoidance experiments with *M. plumulosa* and *N. spinipes* were initiated by seeding one side of the sediment chamber with 30 test organisms. In avoidance assays where contaminated sediment was utilized, *M. plumulosa* and *N. spinipes* were always seeded into the chamber containing contaminated sediment. The removable barrier that divided the overlying water and prevented organism movement between treatments was removed after 10 min, and the organisms were permitted to move around the test chamber as desired. Because the snails had a lower mobility than the amphipods and copepods, the avoidance tests with *P. solida* were initiated by placing 20 snails in a line across the middle of the test chamber directly above the permanent barrier (no removable barrier was necessary).

All avoidance experiments were conducted under ambient conditions in a temperature-controlled laboratory maintained at $21 \pm 1^\circ\text{C}$. For *M. plumulosa* and *P. solida* tests, the overlying water was gently aerated to maintain dissolved oxygen concentrations $>85\%$. Overlying water in *N. spinipes* experiments was not aerated as the bubbling disturbed the sediment. Previous experiments with this species indicated that aeration of the overlying water was not necessary over short time periods (<7 d) to maintain sufficient levels of dissolved oxygen [29]. Subsamples of the overlying water were collected and analyzed for dissolved methods (as described above) every 2 to 3 d during the test and immediately prior to test termination. No food was added to the test chambers for the duration of the test. Tests were terminated by inserting the temporary barriers into the test vessels. The sediment on both sides of the barrier was gently sieved, and the number of animals in the seeded and non-seeded chambers was recorded. For quality-control purposes, an average organism recovery of $\geq 80\%$ was required for a test to be considered acceptable.

General chemistry

New plasticware was used for all chemical analyses. All chemicals used were analytical reagent grade or of equivalent analytical purity. Measurements of pH, salinity, temperature, and dissolved oxygen were made in accordance with the instrument manufacturers' instructions. Deoxygenated waters were prepared by bubbling solutions with high-purity, oxygen-free nitrogen gas for >8 h to give dissolved oxygen concentrations <0.1 mg/L. Measurements of pH, particle size distribution (wet sieving and gravimetry), organic carbon (high-temperature total organic carbon analyzer), and particulate metals (2:1 concentrated HCl:HNO₃, heated) were made as described previously [23,30]. Overlying water samples were rapidly filtered through acid-washed $0.45\text{-}\mu\text{m}$ membrane filters (Minisart; Sartorius) immediately following collection and acidified to 2% HNO₃ (v/v) with concentrated HNO₃ (Tracepure; Merck). Acid-volatile sulfide and simultaneously extracted metals were analyzed according to Simpson [31]. Dissolved metal concentrations in water samples and digested sediments were determined by inductively coupled plasma-atomic emission spectrometry (Varian 730-ES; Varian Australia) calibrated with matrix-matched

Table 1. Properties of uncontaminated sediments

Parameter	A >90% silt	B 50% silt ^a	C 10% silt ^a	D 100% sand
Size <63 μm (%)	98	49.1	9.9	<0.1
AVS ($\mu\text{mol/g}$)	4.5	2.3	0.5	<0.01
Total organic carbon (%)	4.7	2.4	0.6	<0.1
Fe (%)	2.4	1.2	0.3	0.06
Mn ($\mu\text{g/g}$)	71	37	9.8	3
Cu ($\mu\text{g/g}$)	13	7	2.2	1
Pb ($\mu\text{g/g}$)	40	21	5.8	2
Zn ($\mu\text{g/g}$)	210	110	24.3	3
Total PAHs (mg/kg)	5	3.5	2.3	<2
Porewater NH_3 (mg/L)	<5	3	1.4	<1

^a Calculated based on the mixing ratio of >90% silt and 100% sand sediments.

AVS = acid-volatile sulfide; PAH = polycyclic aromatic hydrocarbon.

standards as described in Angel et al. [32]. Analyses of filter and digest blanks, replicates for 20% of samples, analyte sample spikes, and the certified reference material (PACS-2; National Research Council of Canada) were made as part of the quality assurance; and recoveries were within 85 to 110% of expected values. The limits of reporting for the various methods were less than one-tenth of the lowest reported values. All sediment-related concentrations are reported on a dry mass basis.

Data analyses

All statistical analyses were carried out using the Microsoft Excel (2007) data analysis tool pack. The distribution of organisms in avoidance bioassays was reported as a percentage of recovered organisms to account for small differences in the recovery rate of organisms between replicates. The difference in the proportion of organisms on either side of the test chambers was determined by a *t* test, with $p < 0.05$ resulting in a significant difference being detected. In all cases, avoidance was significant when there was a >10% difference in the mean distribution of organisms (between the contaminated and non-contaminated sides of the test vessel) and a lack of overlap of standard errors. Where a difference in the mean distribution of test organisms was <10% (equivalent to a difference of fewer than three animals), it was considered that avoidance had not occurred. This variation was assumed to reflect natural variability in bioassay experiments, and *t* tests were not performed.

RESULTS AND DISCUSSION

Grain size and avoidance behavior in uncontaminated sediment

Sediments can vary greatly in physical and chemical properties, and these variations can influence the nature of the resident ecosystem structure [10,11]. Before the avoidance of *M.*

plumulosa, *N. spinipes*, and *P. solida* was determined for contaminated sediments, we determined whether varying sediment grain size and organic carbon content influenced the behavior of organisms when exposed to uncontaminated sediments (Table 1).

In vessels with uncontaminated silty sediment in both chambers, each of the test species dispersed to create a relatively even distribution of the test organisms on both sides of the test vessel over 48 h (Fig. 1). This confirmed that *M. plumulosa*, *N. spinipes*, and *P. solida* could move freely around the test chambers and had a preference for spreading out rather than remaining in a high-density group. This behavioral characteristic has been shown for several estuarine species including the amphipod *Eohaustorius estuarius* [9] and *Rhepoxynius* spp. [33]. The even distribution of the test organisms also suggests that ambient laboratory conditions did not influence organism dispersal.

When exposed to two different types of uncontaminated sediments *M. plumulosa*, *N. spinipes*, and *P. solida* dispersed throughout the vessel, as they had when contained in vessels with the same sediment on both sides (Fig. 1). For all three species, there was no significant difference in the distribution of the test organisms between the silt reference sediment and the silt and sand sediment mixtures. For *M. plumulosa*, a minor preference for 50% silt over the reference sediment was observed (Fig. 1A). However, a 9.4% difference in the distribution of these organisms between the sediments was considered to be the result of natural variability and not to represent an avoidance response.

The sediment mixtures tested in these experiments encompassed a range of sediment characteristics including particle, acid-volatile sulfide, total organic carbon, and metal concentrations (Table 1). In past studies, estuarine snails (*Ilyanassa*

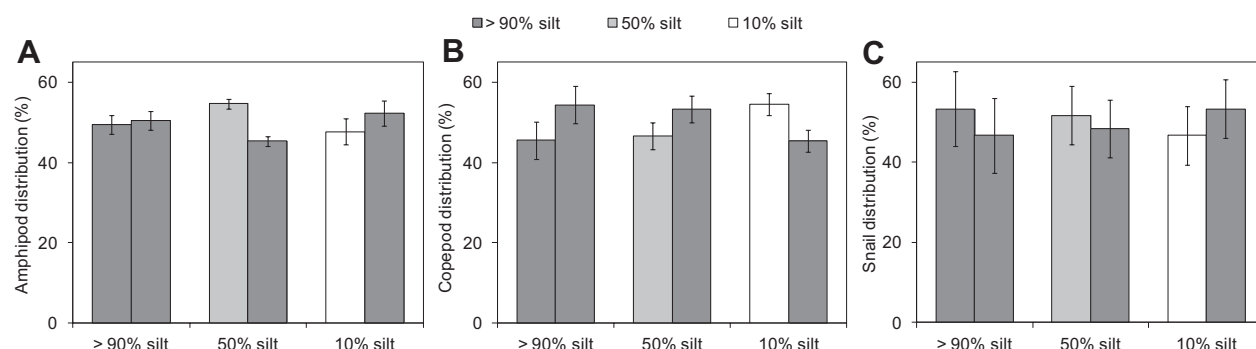


Fig. 1. The distribution of (A) *Melita plumulosa*, (B) *Nitocra spinipes*, and (C) *Phallomedusa solida* when exposed for 48 h to sediments of differing silt composition (>90, 50, and 10% silt; mean $\pm$ standard error, $n = 3$; $n = 4$ for *N. spinipes*). [Color figure can be seen in the online version of this article, available at www.interscience.wiley.com]

Table 2. Sediment chemistry data for the test sediments used in the present study

Parameters ^a	Test Sediment								
	1	2 ^b	3	4	5	6	7	8	9
Silt (%)	76	91	15	79	49	27	47	1	84
AVS (μmol/g)	<5	<5	<5	<5	<5	<5	<5	<5	14 ± 12
TOC (%)	3.9	3.1	5.9	6.8	4.0	3.0	1.5	0.7	6.0
As (μg/g)	3,430	4	7	61	15	ND	18	ND	98
Cd (μg/g)	83	<1	0	19	1	1.1	1	1.1	10
Cu (μg/g)	1,100	6	1,070	108	1,130	71	930	49	112
Pb (μg/g)	1,4400	15	60	830	43	470	36	370	1,730
Zn (μg/g)	1,4500	44	260	2,630	92	1,230	95	1,050	2,800
Amphipod survival (% ± standard error)	12 ± 5	35 ± 30%	24 ± 6	32 ± 2	47 ± 10	59 ± 11	63 ± 4	93 ± 9	96 ± 6

^a Silt refers to percentage of particles <63 μm. All metal concentrations were dilute acid-extractable metals. Amphipod survival was in 10-d toxicity testing.

^b Sediment 2 was a 5% diesel-spiked sediment that contained 26 mg/kg total polycyclic aromatic hydrocarbons (the sum of 16 polycyclic aromatic hydrocarbons [1] and 11,500 mg/kg C10–C36 total petroleum hydrocarbons).

AVS = acid-volatile sulfide. TOC = total organic carbon. ND = not determined.

obsoleta) exposed to coarse sand containing no organic carbon showed a preference for silty sediment. This preference was attributed to the availability of food (as organic matter) in the silty sediment [34,35]. Meadows [36] reported that the amphipod *Corophium volutator* also had a preference for burrowing in finer-grained sediments compared to sandy sediments. The contaminated sediments used in the present study (Table 2) all contained ≥10% silt (except Sediment 8) and had total organic carbon and acid-volatile sulfide levels similar to those tested in the grain size-avoidance study (Fig. 1). Therefore, the properties of the test and reference sediments were not expected to influence the avoidance behavior of *M. plumulosa*, *N. spinipes*, and *P. solida* in subsequent experiments.

Time to response: Dispersal and avoidance of uncontaminated sediment

The time required to observe an avoidance response of *M. plumulosa*, *N. spinipes*, and *P. solida* for contaminated sediment was tested for periods ranging from 3 to 240 h. Under control conditions, *M. plumulosa* distribution was variable for shorter test durations (36 ± 21 to 64 ± 21% at 6 h for seeded and non-seeded sides, respectively). However, for longer durations (24–240 h) an even distribution of organisms occurred (Fig. 2A). This indicated that *M. plumulosa* were actively dispersing and occupying available space in the test vessel within 6 to 24 h but that a 48-h exposure period induced an avoidance response with less variability.

The distribution of *N. spinipes* was also considerably variable within the first 24 h (Fig. 2B), with shorter experiment durations having a greater proportion of organisms remaining in the seeded side of the test vessel (85 ± 4 and 75 ± 4% for 1- and 3-h exposures, respectively). Exposure times of 24 h were sufficient for the copepods to disperse across both sides of the test vessel (Fig. 2B) and were used for the remaining avoidance studies with *N. spinipes*.

The snail *P. solida* dispersed evenly throughout the test vessel after 6 h (48 ± 4% on the seeded side at 6 h; Fig. 2C). Minimal movement was observed in the snails at shorter time periods; therefore, shorter test durations were not included in the remainder of the study. In contrast to the amphipod and copepod, the distribution of *P. solida* between the two chambers of the test vessel did not change significantly as the duration of the experiments was increased.

The difference in time required to achieve an even distribution of the test organisms is expected to be influenced by both the mobility of the species and the starting position at the

beginning of the assay. While *P. solida* was much less mobile than *M. plumulosa* and *N. spinipes* (DJ Ward, personal observation), the shorter dispersal time for *P. solida* is most likely the result of placing them along the central barrier dividing the two

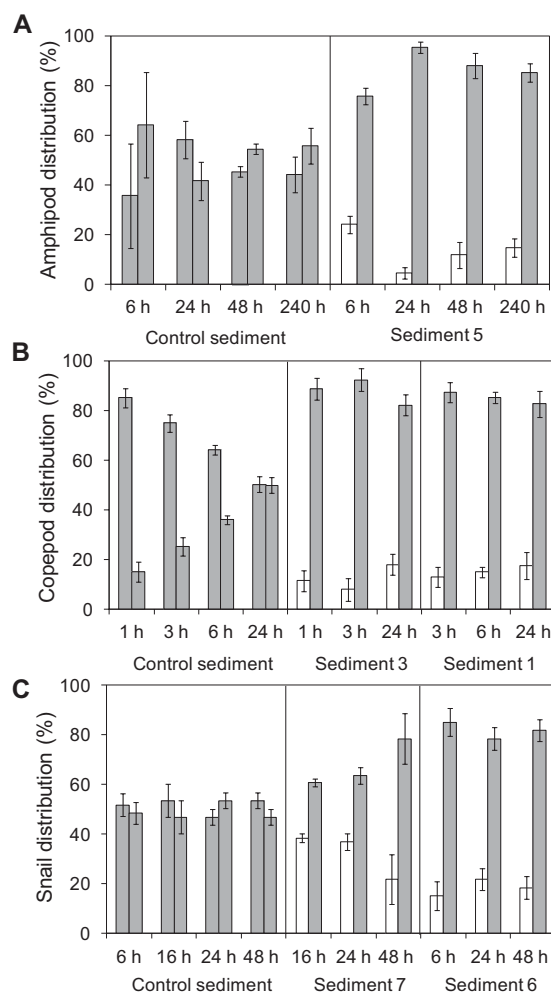


Fig. 2. The observed avoidance response of organisms (mean ± standard error, $n = 3$; $n = 4$ for *N. spinipes*) when exposed to contaminated sediments: (A) *Melita plumulosa* to Sediment 5, (B) *Nitocra spinipes* to Sediments 3 and 1, and (C) *Phallomedusa solida* to Sediments 7 and 6. Bar on the left of each pair represents the seeded side. Filled bars represent control Sediment, and white bars represent contaminated sediment. [Color figure can be seen in the online version of this article, available at www.interscience.wiley.com.]

test chambers. As the copepods and amphipods were distributed throughout the seeded chamber, the organisms potentially had a greater distance to travel than the snails before sensory systems could trigger an avoidance response to the less desirable areas of contaminated sediment.

Zylstra [37] found that dermal sensory cells in the freshwater pulmonate snail *Lymnaea stagnalis* were prevalent on the surface of the tentacles as well as around the lips, the front edge of the foot, and the mantle edge. The concentration of sensory cells on the tentacles, mouth, and margins of the lips reached densities up to 5,000 cells/mm², resulting in a greater number of peripheral sensory neurons than in the central nervous system [38]. In light of this, it makes sense that aquatic snails are sensitive to their surrounding chemical environment and that chemoreception is an important factor controlling the behavior of these organisms. Aquatic pulmonate snails have been shown to be extremely sensitive to chemical cues in the environment. For example, *Physella columbiana* has demonstrated the ability to avoid soil solutions with a mixture of zinc and cadmium at concentrations <10 µg/L [6]. Therefore, we speculate that the chemoreceptors of *P. solida* are more sensitive than those of *N. spinipes* and *M. plumulosa* and, as such, enhance the ability of this species to sense the surrounding environment and move away from unfavorable conditions.

Organism density within a population may be a factor contributing to the speed at which the organisms disperse; for example, Rosenberg et al. [39] found that a higher density of brittle star *Amphiura filiformis* resulted in a significant increase in the rate of dispersal. Estuarine snails have also been shown to exhibit behavioral changes in response to high organism density, including increases in the incidence of floating, emigration, and climbing behavior as well as a reduction in feeding and crawling rates [40,41]. If aquatic organisms prefer low spatial densities, then physically larger species should react to overcrowding more actively to disperse within available space. This behavior was previously observed for two species of estuarine snail which displayed an increase in dispersal among larger (mature) specimens when resources were limited [42]. As *P. solida* was the largest of the test species used in the present study, it is reasonable to expect that population density-related effects may occur and facilitate the dispersal rates. If considering only the dispersal of the test organisms under control conditions, it is possible that a density-dependent effect may encourage the spreading out of organisms in the test vessel; and this may be more significant for *P. solida*. We did not undertake tests to determine whether this may be the case. However, based on the results from the contaminant-avoidance experiments, we found that, under the same density constraints, all three test species showed a preference to migrate away from the contaminated sediment and occupy the clean, habitable sediment despite the resulting higher density of organisms.

Avoidance of contaminated sediment

The ability of the three species to avoid contaminated sediments was tested using the sediments shown in Table 2. The percentage of survival in 10-d lethality tests using the amphipod *M. plumulosa* is also shown. These were field-collected sediments, except for the diesel-spiked Sediment 2 and copper-spiked Sediments 5 and 7.

To determine whether the behavior of the test organisms was altered by the presence of contaminants, contaminated sediments were placed on one side of the test containers and uncontaminated reference sediment on the other. If the organisms no longer dispersed throughout the test vessel but remained

at a higher density on the clean reference sediment, then this would be sufficient evidence that the chemical stimuli (in this case mostly metal contaminants) altered the behavior of the test organisms (Fig. 2).

The times to achieve uniform distribution of *M. plumulosa*, *N. spinipes*, and *P. solida* in uncontaminated sediments under control conditions were 48, 24, and 6 h, respectively; and changes to these times would also be used to provide information on avoidance behavior. We placed *M. plumulosa* in the test vessels with the reference and copper-spiked sediment (Sediment 5). After just 6 h of exposure, the distribution of *M. plumulosa* was significantly lower in the contaminated, seeded side (Sediment 5) compared to the uncontaminated, non-seeded side of the test chamber, with a distribution of 24 ± 4 to $76 \pm 4\%$, respectively (Fig. 2A). Exposure durations of 24, 48, and 240 h showed a similar distribution to the 6-h result, with at least 85% of amphipods residing in the uncontaminated sediment, despite the extended time period. However, significantly more *M. plumulosa* avoided Sediment 5 following a 24-h exposure ($95 \pm 2\%$) than after a 48- or 240-h exposure (88 ± 5 and $85 \pm 4\%$, respectively). Overall, *M. plumulosa* maintained a preference to inhabit clean sediments over longer exposure times, despite a higher population density (number of organisms per given area) than observed under control conditions.

While *M. plumulosa* clearly avoided Sediment 5 after a 6-h exposure period (Fig. 2A), the variability of amphipod distribution at 6 and 24 h in uncontaminated (control) treatments made interpretation of the contaminated sediment results difficult. A longer exposure period of 48 h was chosen for future avoidance experiments as it allowed sufficient time for the organisms in the control sediments to disperse and interact with the sediment before the experiment was terminated.

The copepod *N. spinipes* had the fastest avoidance time of the three test species. When exposed to contaminated Sediments 1 and 3 (Table 2) >87% of copepods occupied the non-contaminated chamber of the test vessel after only 3 h of exposure (Fig. 2B). This is a highly significant proportion of the copepod population migrating away from the contaminated sediment, showing a clear preference for the non-contaminated sediment. For Sediment 3, 1-h exposure resulted in $89 \pm 4\%$ of copepods moving to the uncontaminated sediments in the test chamber. For both sediments, there was no significant difference between the number of avoiding organisms at the termination of the experiment for all exposure times; hence, the increased exposure duration did not elicit a greater avoidance effect.

When exposed to Sediment 7, *P. solida* had an avoidance time of ≥ 16 h, with a significant number of snails ($61 \pm 2\%$) avoiding the contaminated sediment (Fig. 2C). This avoidance increased over time, with 48 h resulting in $78 \pm 10\%$ of *P. solida* migrating to the uncontaminated sediment. When exposed to Sediment 6, avoidance occurred after a shorter time than for Sediment 7, with $85 \pm 6\%$ of snails migrating away from the contaminated Sediment after 6 h.

The avoidance response time of *P. solida* was similar to the 6- to 24-h avoidance time of the estuarine snail *Ilyanassa obsoleta* when exposed to clean and contaminated harbor sediments [34,35]. Despite extending the duration of exposure to as much as 48 h, *P. solida* maintained a preference for the control sediment. Overall, the results show that in the presence of an unfavorable chemical stimulus, all three species will migrate away from the source and will remain there for extended periods of time.

Validation of sediment avoidance of *M. plumulosa*

The amphipod *M. plumulosa* is commonly used in Australia for sediment quality assessment [18,30]. For nine contaminated sediments that caused varying degrees of toxicity to *M. plumulosa* in 10-d lethality tests (Table 2), the avoidance response after 48 h was determined (Fig. 3).

Six of the contaminated sediments resulted in a significant avoidance response in *M. plumulosa* within 48 h. The greatest avoidance occurred for Sediments 1, 2, and 3, which resulted in 97, 94, and 93% of *M. plumulosa* moving to the uncontaminated sediment, respectively. These sediments were highly toxic, with Sediments 1 and 3 resulting in >75% 10-d lethality (Table 2). While it was not possible to accurately determine the 10-d lethality of the diesel-spiked Sediment 2 (perhaps due to the volatility of the diesel changing during the exposure), the sediment was strongly avoided. Exposure of *M. plumulosa* to sediments that were moderately toxic (Sediments 5, 6, and 7) did not always result in a significant avoidance response. More than 75% of amphipods avoided Sediments 5 and 7; however, avoidance was not significant for Sediment 6 despite resulting in >40% lethality in 10-d toxicity tests. Sediments 5 and 7 contained similar concentrations of copper but differed significantly in grain size distribution (Table 2). When *M. plumulosa* was exposed to sediments that did not elicit acute toxicity (Sediments 8 and 9), avoidance was not observed.

In all exposure chambers the overlying water was continuously exchanged between the two sides and dissolved metal concentrations in the overlying waters were relatively low (e.g., <30 µg Cu/L for tests with sediments 5 and 7) and not significantly different for samples taken from the different sides of the test chamber (data not shown). As it is known that *M. plumulosa* is sensitive to sediment-bound copper [28,43], avoidance of these sediments was believed to be due to differences in particulate copper or the flux of copper at the sediment–water interface, rather than differences in dissolved copper in the overlying water. Sediments 5 and 7 were copper-spiked sediments, and although they had been equilibrated and pore-water copper concentrations were <5 µg/L (unpublished results), the copper in these sediments was likely to be more bioavailable than if present in field-collected sediments with similar copper concentrations [30]. The greater avoidance of Sediments 5 and 7 compared to Sediment 4, which had high lead

and zinc concentrations and greater toxicity, indicates that *M. plumulosa* may be more sensitive to copper than other metal contaminants. However, although dilute acid-extractable metal concentrations (Table 2) are expected to predict the bioavailability of metals better than total metal concentrations [25], the fact that Sediments 5 and 7 were copper-spiked may have resulted in this copper being easier for *M. plumulosa* to sense than the lead and zinc in the field-collected Sediment 4.

Despite the high lead and zinc concentrations in Sediment 9, this sediment was not avoided and was also not toxic to *M. plumulosa*. This sediment had a relatively high concentration of acid-volatile sulfide (14 ± 12 µmol/g), and the lead and zinc are predicted to be present in sulfide phases of low bioavailability [44]. Interestingly, *M. plumulosa* appeared to show a slight attraction to this sediment, with 61% of amphipods remaining on the seeded portion of the test vessel, effectively choosing to inhabit the contaminated sediment. However, there was no statistical significance between the distribution of amphipods on the seeded and unseeded chambers at the end of the test.

Experiments with uncontaminated sediments found that sediment grain size did not influence the avoidance response for *M. plumulosa*, and this also appeared to be the case for the contaminated sediments. Marklevitz et al. [35] found that avoidance of contaminated harbor sediments by a mollusk was a balance between attraction to food and aversion to contamination. Given the high concentration of organic carbon present in Sediments 4 and 9 (6.8 and 6.0%, respectively, which exceeds the reference sediment), it is possible that the attraction to the added nutrition of the test sediments (presumably provided by the additional organic content of the sediment) may have affected the observed avoidance of these sediments. Sediment 3 also had a high organic carbon content (5.9%); however, the observation that it was both avoided and toxic (>75% mortality) indicates that if nutrition in the form of total organic carbon influences avoidance behavior, then the quality of the total organic carbon may also be important. The major contaminant in Sediment 3 was copper and, when considering the results for Sediments 5 and 7, may indicate that sediment copper is more readily detected and avoided by the amphipod.

Influence of hazard magnitude on avoidance behavior

The increasing hazard posed by contaminated sediments is thought to elicit a faster or greater response from exposed organisms [35,45]. For *M. plumulosa*, the results of the present study indicate a similar trend as the degree of avoidance was generally greater for sediment which elicited greater toxicity (Table 2, Fig. 3).

This indicates that the hazard imposed by the contaminated sediment (i.e., toxicity) influences the number of amphipods that avoid the unfavorable sediment over a 48-h period. The avoidance response of the copepods to Sediments 1 and 3 was very similar to the amphipod response (Fig. 2), but over a shorter time, with 3 h (87 ± 4 and $92 \pm 5\%$, respectively) and 24 h (83 ± 5 and $82 \pm 4\%$, respectively). Given the similar toxicity of these two sediment samples and the similarity between the avoidance response of the copepods and amphipods, the avoidance behavior of *N. spinipes* may also increase as toxicity increases. However, overall an insufficient range of contaminated sediments was tested to determine if this was true for *N. spinipes* and *P. solidus*.

Comparing the distribution of *M. plumulosa* and *N. spinipes* exposed to control conditions at each time point and the corresponding contaminant avoidance provides strong evidence

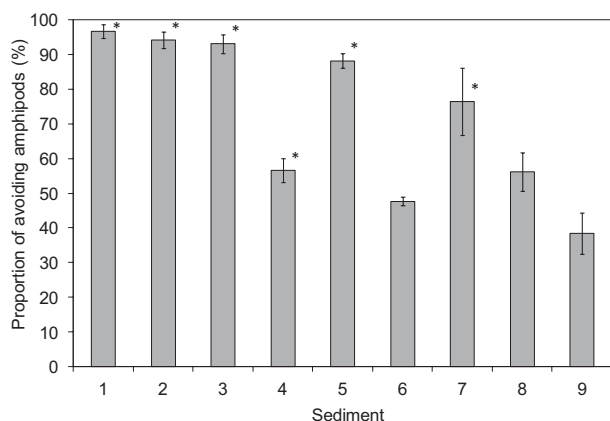


Fig. 3. The observed avoidance response of *Melita plumulosa* following 48-h exposure to field-contaminated and laboratory-spiked sediments (mean $\pm$ standard error, $n=3$). *Significant avoidance of contaminated sediment. The 10-d whole sediment toxicity decreases from left to right. [Color figure can be seen in the online version of this article, available at wileyonlinelibrary.com]

that the presence of sediment-bound contaminants influences the observed avoidance response (Fig. 2). In test chambers containing only control sediment, approximately 24 h was required for a significant percentage of the organisms to migrate away from the seeded portion of the container compared to <6 h when contaminants were present. This was despite possible density-related pressures that may be placed on the organisms due to seeding on only one side of the test container. It is particularly evident for *N. spinipes*, where after 1 h only $15 \pm 4\%$ of the copepods migrated away from the seeded control treatment compared to $89 \pm 4\%$ of copepods avoiding Sediment 3 (onto which they were seeded) in the same period of time. While the response was not as rapid or clear for *M. plumulosa*, the distribution after 6 h was highly variable for the controls; but $76 \pm 4\%$ of amphipods avoided the contaminated Sediment 5. The enhanced activity of *M. plumulosa* and *N. spinipes* was clearly avoidance behavior, and toxicity would have occurred if the species remained in the sediments.

The snail *P. solida* was slower-moving, but more rapid avoidance was observed for Sediment 6 (≤ 6 h) than for Sediment 7 (~ 16 h; Fig. 2). While Sediments 6 and 7 caused similar acute toxicity (Table 2), accompanying studies showed that Sediment 6 caused much greater chronic toxicity to *M. plumulosa* (D Spadaro, CSIRO Land and Water, Lucas Heights, NSW, Australia, unpublished data). In addition, the degree of avoidance exhibited by *P. solida* exposed to Sediment 6 for 6, 24, or 48 h was not significantly different ($< 7\%$ difference in the number of avoiding organisms). This suggests that density-related pressures did not influence the behavior of *P. solida*; however, tests were not undertaken to determine whether this was the case here. After 48 h, the snails preferentially maintained a higher density in clean sediments rather than inhabiting the contaminated Sediments 6 and 7.

Previous studies have demonstrated behavioral changes in benthic organisms resulting from the presence of environmental contaminants. The bivalve *Macomona liliana* was shown to crawl or drift away from sediments spiked with copper and zinc [14,46]. Thus, a behavioral endpoint for the bivalve was more sensitive than both burial/morbidity and mortality endpoints [46]. *Physella columbiana*, an aquatic pulmonate snail, demonstrated the ability to detect cadmium and zinc at concentrations of $9 \mu\text{g Cd/L}$, $56 \mu\text{g Zn/L}$, and below $10 \mu\text{g/L}$ for mixtures [6]. Oakden [33] reported the avoidance of metal- and sewerage-enriched sediments by phoxocephalid amphipods (*Rhepoxynius* spp.), while Kravitz et al. [9] found that *Eohaustorius estuarii* could avoid sediments with moderate to heavy polycyclic aromatic hydrocarbon contamination.

CONCLUSIONS

The present study demonstrated the ability of *M. plumulosa*, *N. spinipes*, and *P. solida* to respond to chemical cues in the environment and choose a habitat that provides the best opportunity for survival by avoiding contaminated sediment. The avoidance response of the test organisms indicates that static 10-d toxicity methods are likely to overestimate toxicity for species, which would avoid contamination in heterogeneous field settings. The sensitivity of the avoidance response of each species to sediment-bound contaminants indicates that they may be suitable for development as rapid screening methods to assess sediment quality. The study provided strong evidence that the avoidance response of *M. plumulosa* was related to the toxicity associated with the sediment. For this species the 48-h avoidance endpoint correctly identified as a potential hazard six

of seven sediment samples that caused significant acute toxicity during 10-d exposures.

SUPPLEMENTAL DATA

Figure S1. (10 KB PDF).

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REFERENCES

- Cairns J. 1992. The threshold problem in ecotoxicology. *Ecotoxicology* 1:3–16.
- Atchison GJ, Sandheinrich MB, Bryan MD. 1996. Effects of environmental stressors on interspecific interactions of aquatic animals. In Newman MC, Jagoe CH, eds *Ecotoxicology: A Hierarchical Treatment*. Lewis, Boca Raton, FL, USA, pp 319–337.
- Amiard-Triquet C. 2009. Behavioral disturbances: The missing link between sub-organismal and supra-organismal responses to stress? Prospects based on aquatic research. *Hum Ecol Risk Assess* 15:87–110.
- Weber DN. 1997. Mechanisms of behavioral toxicology: An integrated approach. *Am Zool* 37:343–345.
- Roast S, Widdows J, Jones M. 2000. Disruption of swimming in the hyperbenthic mysid *Neomysis integer* (Peracarida: Mysidacea) by the organophosphate pesticide chlorpyrifos. *Aquat Toxicol* 47:227–241.
- Lefcort H, Abbott D, Cleary D, Howell E, Keller N, Smith M. 2004. Aquatic snails from mining sites have evolved to detect and avoid heavy metals. *Arch Environ Contam Toxicol* 46:478–484.
- Lopes I, Baird DJ, Ribeiro R. 2004. Avoidance of copper contamination by field populations of *Daphnia longispina*. *Environ Toxicol Chem* 23:1702–1708.
- Riddell DJ, Culp JM, Baird DJ. 2005. Behavioral responses to sublethal cadmium exposure within an experimental aquatic food web. *Environ Toxicol Chem* 24:431–441.
- Kravitz MJ, Lamberson JO, Ferraro SP, Swartz RC, Boese BL, Specht DT. 1999. Avoidance response of the estuarine amphipod *Eohaustorius estuarii* to polycyclic aromatic hydrocarbon contaminated, field collected sediments. *Environ Toxicol Chem* 18:1232–1235.
- Chariton AA, Roach AC, Simpson SL, Batley GE. 2010. Influence of the choice of physical and chemistry variables on interpreting patterns of sediment contaminants and their relationships with estuarine macrobenthic communities. *Mar Freshw Res* 61:1109–1122.
- Rakocinski CF, Brown SS, Gaston GR, Heard RW, Walker WW, Summers JK. 1997. Macrobenthic responses to natural and contaminant-related gradients in northern Gulf of Mexico estuaries. *Ecol Appl* 7:1278–1298.
- Hellou J. 2011. Behavioural ecotoxicology, an “early warning” signal to assess environmental quality. *Environ Sci Pollut Res* 18:1–11.
- Exley C. 2000. Avoidance of aluminum by rainbow trout. *Environ Toxicol Chem* 19:933–939.
- Roper DS, Nipper MG, Hickey CW, Martin ML, Weatherhead MA. 1995. Burial, crawling and drifting behaviour of the bivalve *Macomona liliana* in response to common sediment contaminants. *Mar Pollut Bull* 31:471–478.
- Eriksson Wiklund AK, Borjesson T, Wiklund SJ. 2006. Avoidance response of sediment living amphipods to zinc pyriothione as a measure of sediment toxicity. *Mar Pollut Bull* 52:96–99.
- Burton GA, Pitt R, Clark S. 2000. The role of traditional and novel toxicity test methods in assessing stormwater and sediment contamination. *Crit Rev Environ Sci Technol* 30:413–447.
- American Society for Testing and Materials. 2003. Standard test method for measuring the toxicity of sediment-associated contaminants with estuarine and marine invertebrates. E1367-03e1. Philadelphia, PA.
- Simpson S, Batley G, Chariton A, Stauber J, King C, Chapman J, Hyne R, Gale S, Roach A, Maher W. 2005. *Handbook for Sediment Quality Assessment*. Environmental Trust, Canberra, ACT, Australia.
- Kennedy AJ, Steevens JA, Lotufo GR, Farrar JD, Reiss MR, Kropp RK, Doi J, Bridges TS. 2009. A comparison of acute and chronic toxicity methods for marine sediments. *Mar Environ Res* 68:118–127.
- Egeler P, Gilberg D, Fink G, Duis K. 2010. Chronic toxicity of ivermectin to the benthic invertebrates *Chironomus riparius* and *Lumbriculus variegatus*. *J Soils Sediments* 10:368–376.
- U.S. Environmental Protection Agency. 2001. Methods for assessing the chronic toxicity of marine and estuarine sediment-associated

- contaminants with the amphipod *Leptocheirus plumulosus*. EPA 600/R-01/020. Washington, DC.
22. Roach AC, Lim RP. 2000. Variation in the population dynamics of the intertidal pulmonate gastropod *Salinator solida* Martens (Gastropoda: Amphibolidae) at Towra Point, NSW, Australia. *Wetl Ecol Manag* 8: 53–69.
 23. Spadaro D, Micevska T, Simpson S. 2008. Effect of nutrition on toxicity of contaminants to the epibenthic amphipod *Melita plumulosa*. *Arch Environ Contam Toxicol* 55:593–602.
 24. Perez-Landa V, Simpson SL. 2011. A short life-cycle test with the epibenthic copepod *Nitocra spinipes* for sediment toxicity assessment. *Environ Toxicol Chem* 30:1430–1439.
 25. Simpson SL, Spadaro DA. 2011. Performance and sensitivity of rapid sublethal sediment toxicity tests with the amphipod *Melita plumulosa* and copepod *Nitocra spinipes*. *Environ Toxicol Chem* 30: 2326–2334.
 26. Simpson SL, Burston VL, Jolley DF, Chau K. 2006. Application of surrogate methods for assessing the bioavailability of PAHs in sediments to a sediment ingesting bivalve. *Chemosphere* 65:2401–2410.
 27. Simpson SL, Angel BM, Jolley DF. 2004. Metal equilibration in laboratory-contaminated (spiked) sediments used for the development of whole-sediment toxicity tests. *Chemosphere* 54:597–609.
 28. King CK, Gale SA, Hyne RV, Stauber JL, Simpson SL, Hickey CW. 2006. Sensitivities of Australian and New Zealand amphipods to copper and zinc in waters and metal-spiked sediments. *Chemosphere* 63:1466–1476.
 29. Ward DJ, Perez-Landa V, Spadaro DA, Simpson SL, Jolley DF. 2011. An assessment of three harpacticoid copepod species for use in ecotoxicological testing. *Arch Environ Contam Toxicol* 61:414–425.
 30. Strom D, Simpson SL, Batley GE, Jolley DF. 2011. The influence of sediment particle size and organic carbon on toxicity of copper to benthic invertebrates in oxic/suboxic surface sediments. *Environ Toxicol Chem* 30:1599–1610.
 31. Simpson SL. 2001. A rapid screening method for acid-volatile sulfide in sediments. *Environ Toxicol Chem* 20:2657–2661.
 32. Angel BM, Simpson SL, Jolley DF. 2010. Toxicity to *Melita plumulosa* from intermittent and continuous exposures to dissolved copper. *Environ Toxicol Chem* 29:2823–2830.
 33. Oakden JM. 1984. Behavioural responses of a phoxocephalid amphipod to organic enrichment and trace metals in sediment. *Mar Ecol Prog Ser* 14:253–257.
 34. Marklevitz S, Almeida E, Flemming J, Hellou J. 2008. Determining the bioavailability of contaminants and assessing the quality of sediments. Part 1: Variables affecting the behavioural response of *Ilyanassa obsoleta* toward contaminated sediments. *J Soils Sediments* 8:86–91.
 35. Marklevitz S, Almeida E, Flemming J, Hellou J. 2008. Determining the bioavailability of contaminants and assessing the quality of sediments. Part 2: Behavioural response of snails *Ilyanassa obsoleta* toward contaminated harbour sediments. *J Soils Sediments* 8:92–97.
 36. Meadows PS. 1964. Substrate selection by corophium species: The particle size of substrates. *J Anim Ecol* 33:387–394.
 37. Zylstra U. 1971. Distribution and ultrastructure of epidermal sensory cells in the freshwater snails *Lymnaea stagnalis* and *Biomphalaria pfeifferi*. *Neth J Zool* 22:283–298.
 38. Wyeth RC, Croll RP. 2011. Peripheral sensory cells in the cephalic sensory organs of *Lymnaea stagnalis*. *J Comp Neurol* 519:1894–1913.
 39. Rosenberg R, Nilsson HC, Hollertz K, Hellman B. 1997. Density-dependant migration in an *Amphiura filiformis* (Amphiuridae, Echinodermata) infaunal population. *Mar Ecol Prog Ser* 159:121–131.
 40. Levinton JS. 1979. The effect of density upon deposit-feeding populations: Movement, feeding and floating of *Hydrobia ventrosa* Montagu (Gastropoda: Prosobranchia). *Oecologia* 43:27–39.
 41. Lopez-Figueroa F, Niell FX. 1987. Feeding behaviour of *Hydrobia ulvae* (Pennant) in microcosms. *J Exp Mar Biol Ecol* 114:153–167.
 42. Byers JE. 2000. Effects of body size and resource availability on dispersal in a native and non-native estuarine snail. *J Exp Mar Biol Ecol* 248:133–150.
 43. Gale SA, King CK, Hyne RV. 2006. Chronic sublethal sediment toxicity testing using the estuarine amphipod, *Melita plumulosa* (Zeidler): Evaluation using metal spiked and field contaminated sediments. *Environ Toxicol Chem* 25:1887–1898.
 44. U.S. Environmental Protection Agency. 2005. Procedures for the derivation of equilibrium partitioning sediment benchmarks (ESBs) for the protection of benthic organisms: Metal mixtures (cadmium, copper, lead, nickel, silver, and zinc). EPA 600/R-02/011, Washington, DC.
 45. De Lange H, Sperber V, Peeters E. 2006. Avoidance of polycyclic aromatic hydrocarbon-contaminated sediments by the freshwater invertebrates *Gammarus pulex* and *Asellus aquaticus*. *Environ Toxicol Chem* 25:452–457.
 46. Roper D, Hickey C. 1994. Behavioural responses of the marine bivalve *Macomona liliana* exposed to copper- and chlordane-dosed sediments. *Mar Biol* 118:673–680.